

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 1 – Case Study (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO3	Assessment:	Assessment 1 – Case Study: Convolutional Neural Networks
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I T Bhupal/192425243 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: bhupalreddy.

QUESTIONS: 5

Total Marks: 50

Annexure A

Case Study: Convolutional Neural Networks

Case Study

A healthcare organization is planning to develop an AI-based Chest X-Ray Analysis System to assist doctors in detecting pneumonia. The system will use a large dataset of chest X-ray images labeled as Normal and Pneumonia. A CNN model will be trained to automatically learn important visual patterns from the images and classify new X-ray images.

Based on the above case study, answer the following questions:

1. Selection of CNN

Explain why a Convolutional Neural Network (CNN) would be suitable for developing the proposed chest X-ray classification system. Compare its approach to traditional machine learning methods that require manual feature extraction. Discuss how CNNs can automatically learn spatial and visual features from X-ray images.

2. Convolution Layer and Filters

Explain how the convolution layer processes a chest X-ray image. Describe the role of filters/kernels in detecting important patterns such as edges, textures, shapes, and other visual abnormalities. Explain how different filters can learn different features during CNN training.

3. Feature Maps

After applying multiple filters to an input X-ray image, the CNN produces several feature maps. Explain what feature maps represent and how the information captured by feature maps changes as the image passes through deeper CNN layers. Give examples of simple and complex features that may be detected while identifying pneumonia.

4. Pooling Layer

Explain the purpose of the pooling layer in the proposed CNN architecture. Compare Max Pooling and Average Pooling in terms of their working principles and effect on feature maps. Considering the chest X-ray application, explain which pooling technique you would prefer and justify your choice.

5. Fully Connected Layer

After several convolution and pooling operations, the extracted features need to be converted into a final prediction. Explain the role of the Fully Connected (FC) layer in the CNN architecture. How does it use the features extracted from the X-ray to classify the image into Normal or Pneumonia?

6. Parameter Sharing

CNNs use the concept of parameter sharing in convolution layers. Explain how parameter sharing works and why it is important when processing high-resolution medical images. Discuss how parameter sharing reduces the number of trainable parameters compared with a fully connected neural network.

7. Overfitting and Regularization

Suppose the CNN achieves 98% accuracy on the training dataset but only 78% accuracy on unseen X-ray images. Identify the problem occurring in the model and explain why it happens. Discuss suitable regularization techniques, such as Dropout, Data Augmentation, and L1/L2 regularization, that could be used to improve the model's generalization ability.

8. AlexNet Architecture

The development team is considering AlexNet as one of the CNN architectures for the project. Explain the major architectural features of AlexNet, including convolution layers, pooling layers, activation functions, and fully connected layers. Discuss how AlexNet contributed to the development and popularity of deep learning for image classification.

9. AlexNet vs ResNet

The organization is also considering ResNet instead of AlexNet. Compare AlexNet and ResNet based on their architecture, depth, parameter efficiency, training difficulty, and ability to learn complex features. Based on these differences, recommend which architecture would be more appropriate for the chest X-ray classification problem and provide suitable justification.

10. CNN Applications

CNNs are widely used beyond medical image analysis. Explain at least three real-world applications of CNNs, such as facial recognition, autonomous vehicles, object detection, security surveillance.

1. Selection of CNN

Suitability for Chest X-ray Classification: A Convolutional Neural Network (CNN) is highly suitable for analyzing chest X-rays because it is specifically designed to process two-dimensional grid data, like images. Unlike traditional artificial neural networks that flatten an image into a 1D line of pixels—destroying the spatial relationships between them—a CNN preserves the 2D spatial structure. This is critical in medical imaging, where the location and shape of an abnormality (like fluid in the lungs) dictate the diagnosis.

Comparison to Traditional Machine Learning: Traditional machine learning approaches (such as Support Vector Machines or Random Forests) require manual feature extraction. A human engineer must mathematically define patterns, edges, and textures before feeding the data into the model. For complex medical images, defining exactly what pneumonia looks like mathematically is incredibly difficult. CNNs bypass this limitation by automatically learning to extract the most relevant features directly from the raw pixel data during the training phase.

Learning Spatial and Visual Features: CNNs learn automatically through layers of convolutions and a process called backpropagation. As the network is fed thousands of labeled X-rays, it continuously adjusts its internal mathematical weights. Over time, the network autonomously figures out which visual patterns (like lung opacities) correlate with pneumonia, without requiring human intervention to define those features.

2. Convolution Layer and Filters

Processing the X-ray Image: The convolution layer is the core building block of the CNN. It processes the chest X-ray by sliding a small grid of weights (called a window or filter) systematically across the entire image. At every step, it performs a mathematical operation (a dot product) between the filter's weights and the pixels it is hovering over.

The Role of Filters/Kernels: Filters act as highly specific pattern detectors. When a filter slides over a section of the X-ray that matches its specific pattern, it produces a high mathematical output. This allows the network to detect important visual abnormalities such as sharp edges, unique textures, and distinct shapes.

How Different Filters Learn Different Features: A single convolution layer uses many different filters simultaneously. During the training process, the network optimizes these filters to look for different things. For example, one filter's weights might adjust to exclusively detect vertical lines (identifying the ribcage), while another filter might learn to detect soft, cloudy textures (identifying fluid or inflammation).

3. Feature Maps

What Feature Maps Represent: A feature map is the direct mathematical output generated when a filter slides across the input image. It is essentially a new, filtered version of the original X-ray that acts as a "heat map," visually highlighting exactly where a specific pattern or feature was found by that filter.

Changes in Deeper Layers: As the image passes through deeper layers of the CNN, the feature maps undergo a transformation. In the early (shallow) layers, feature maps retain a high spatial resolution and look very similar to the original image. As they progress deeper, they shrink in height and width but increase in depth (the number of abstract concepts detected). They transition from representing raw visual pixels to representing high-level, abstract semantic concepts.

Examples of Simple vs. Complex Features:

- **Simple Features (Early Layers):** The network detects low-level patterns such as stark contrasting edges (the border between the lung wall and air) or simple curves (the shape of the diaphragm).
- **Complex Features (Deep Layers):** The network combines the simple features to detect high-level medical anomalies, such as patchy white infiltrates, cloud-like consolidations in the lower lobes, or blurred heart borders that are indicative of pneumonia.

4. Pooling Layer

Purpose of the Pooling Layer: The pooling layer is used between convolution operations to perform dimensionality reduction. It shrinks the height and width of the feature maps, which drastically reduces the computational load and the number of parameters the network has to process. Additionally, it makes the model "spatially invariant," meaning the network can still accurately detect pneumonia even if the patient's lungs are slightly off-center or rotated in the X-ray.

Max Pooling vs. Average Pooling:

- **Max Pooling:** Slides a window across the feature map and retains only the *maximum* pixel value within that region, discarding the rest. This forcefully highlights the most prominent, intense features.
- **Average Pooling:** Calculates the *mean* value of all pixels in the window. This results in a smoother, diluted representation of the features.

Preferred Technique and Justification: For chest X-ray classification, **Max Pooling** is the preferred technique. In an X-ray, healthy, air-filled lungs appear dark, while pneumonia presents as bright white, hyperdense opacities. Max pooling acts as a signal amplifier; it ensures that these high-intensity, bright abnormalities are preserved and prioritized as the data shrinks, while the darker, less relevant background tissue is filtered out.

5. Fully Connected Layer

Role in the CNN Architecture: The Fully Connected (FC) layer serves as the final decision-making component—or the classifier—of the Convolutional Neural Network. While the preceding convolution and pooling layers function as "feature extractors" to identify spatial patterns, the FC layer takes all of these extracted patterns and translates them into a final diagnosis.

How it Classifies the X-ray (Normal vs. Pneumonia):

- **Flattening the Data:** Before the data can enter the FC layer, the output from the final pooling layer (which is a series of 2D feature maps) must be "flattened." This process unrolls the grid-like visual data into a single, continuous 1D vector (a long list of numbers).
- **Global Connections:** This flattened vector is fed into the FC layer, which operates like a traditional neural network. In this layer, every single neuron is mathematically connected to every single feature extracted from the X-ray.
- **Learning Complex Correlations:** The FC layer learns the complex, non-linear relationships between the detected visual features. For example, the network might learn that the simultaneous presence of a "cloudy texture" feature and a "blurred lower lung contour" highly correlates with an active infection.
- **Final Prediction Output:** The last stage of the FC layer is the output node. For this binary classification task, it uses an activation function like Sigmoid to weigh all the combined features and calculate a final probability score between 0 and 1

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____